# Equivalent expressions — Part 1 answer key

| # | ID | Answer | Difficulty |
|---|---|---|---|
| 1.1 | 04639-DC | C | Easy |
| 1.2 | be5faa6c-0169-41dd-90d9-1eb1a8bdeca1 | D | Easy |
| 1.3 | 21bb428c-50de-4b82-ba02-72c7b6db2d11 | A | Easy |
| 1.4 | 440a7343-d56b-4b84-936d-ec6c5596d326 | A | Easy |
| 1.5 | a81bca68-7fa7-4265-906c-6d5f61769b51 | C | Easy |
| 1.6 | 14779160-e23c-403b-98af-36fdefea4b8e | B | Easy |
| 1.7 | f0feb4ae-ee24-4b97-b265-104550acbce4 | B | Easy |
| 1.8 | 52214fdf-56c8-448c-ac5e-c62e616b3288 | B | Easy |
| 1.9 | 029332-DC | D | Easy |
| 1.10 | 1620a4b4-e1d2-49e7-aae0-096ea9b959f1 | A | Easy |
| 1.11 | 016559-DC | A | Easy |
| 1.12 | 4aae6d4f-125f-4c81-812c-7d232ae7986c | B | Easy |
| 1.13 | 022215-DC | D | Easy |
| 1.14 | 05476-DC | A | Easy |
| 1.15 | 9c3d170a-52ba-4266-b573-bfa73fbbe581 | D | Easy |
| 1.16 | 82d332a7-36d9-4422-9783-754cdaf665a2 | B | Easy |
| 1.17 | 11aa8904-6d75-4f09-b150-a48f09955868 | D | Easy |
| 1.18 | 8a43ff56-7c88-420c-8b4c-bc730b55e931 | D | Easy |
| 1.19 | 4331067c-b0f9-4aaf-a21b-3d660144cf68 | B | Easy |
| 1.20 | 031616-DC | D | Easy |
| 1.21 | 027693-DC | A | Easy |
| 1.22 | 4363efa2-3674-407a-a249-7f0cf0daff53 | A | Easy |

| | | | |
|---|---|---|---|
| 1.23 | 885afb10-2178-4564-84ab-6e185b004a41 | B | Easy |
| 1.24 | a7c54228-4ce0-4f08-8ae4-811ea92339af | C | Easy |
| 1.25 | 017096-DC | D | Easy |
| 1.26 | 70a9ddd5-661b-45e3-b279-9c6de073b79d | A | Easy |
| 1.27 | 707e44df-b9c2-476c-a030-957fe31a519a | D | Easy |
| 1.28 | 56e878c9-7295-480c-8ba1-5c2246df36be | A | Easy |
| 1.29 | 5d93e88d-9666-4e29-baab-4dcd15423597 | B | Easy |
| 1.30 | 238bf570-e789-4f6b-9588-2ff4572b2368 | C | Easy |
| 1.31 | e31accbd-71ed-4323-82c0-b30186a84861 | B | Easy |
| 1.32 | e2375cf4-517a-4d38-94ef-5c6c3cc7b879 | B | Easy |
| 1.33 | 8ef4c8a0-31ce-4c6e-b0e5-d7b0b944b67d | A | Easy |
| 1.34 | 295f9556-4a27-49ca-b269-fb8917b7e050 | C | Easy |
| 1.35 | ba751bc0-12b0-41ac-9d53-b1375e5b11f4 | D | Easy |
| 1.36 | 025717-DC | C | Easy |
| 1.37 | 79ab98cb-39f9-474f-8017-4a2677075b32 | D | Easy |
| 1.38 | 027123-DC | B | Easy |
| 1.39 | 7a034397-6081-46d4-b5f6-0a2735b3a604 | D | Easy |
| 1.40 | 8dbb28ea-e265-4e97-95ca-ce81e767f4e8 | B | Easy |
| 1.41 | 028884-DC | B | Easy |
| 1.42 | ff1e479f-7df1-4306-a31d-d9354d96102e | C | Easy |
| 1.43 | d3f7c3fc-9ac9-456f-8bf8-53653016b0fb | D | Easy |
| 1.44 | 025800-DC | D | Medium |
| 1.45 | d469dd2b-510b-49e3-b521-1271ecbbfa1e | A | Medium |
| 1.46 | afeb4dfe-3156-4234-af28-242e9b1500f8 | 9 | Medium |
| 1.47 | f4365bf0-90a1-4025-b4b6-b2ba9e88b93a | B | Medium |
| 1.48 | 09404-DC | B | Medium |

| | | | |
|---|---|---|---|
| 1.49 | 34c2743d-443b-43f9-ad72-c43601418315 | B | Medium |
| 1.50 | 63296c8a-e373-4a99-89f1-f1ceaa55e479 | A | Medium |
| 1.51 | 027180-DC | ? | Medium |
| 1.52 | ba562150-ace0-4dca-b1c3-c7c1019ccee7 | C | Medium |
| 1.53 | 322bacf8-a04d-4c31-8a29-228d7b8ccca3 | C | Medium |
| 1.54 | 025483-DC | A | Medium |
| 1.55 | 4544da8e-b007-4229-a855-ce10c2bbf557 | B | Medium |
| 1.56 | 2f0bba12-b9f1-460c-b120-0adaf38cf4db | B | Medium |
| 1.57 | 026581-DC | D | Medium |
| 1.58 | 37e39587-24a9-4c5a-b541-d42a0a10ed7a | B | Medium |

## Rationales

▼ 1.1 — C

Choice C is correct. Using the distributive property to multiply 3 and $(x+5)$ gives $3x+15-6$, which can be rewritten as $3x+9$.

Choice A is incorrect and may result from rewriting the given expression as $3(x+5-6)$.
Choice B is incorrect and may result from incorrectly rewriting the expression as $(3x+5)-6$.
Choice D is incorrect and may result from incorrectly rewriting the expression as $3(5x)-6$.

▼ 1.2 — D

Choice D is correct. The given expression follows the difference of two squares pattern, $x^2 - y^2$, which factors as $x - yx + y$. Therefore, the expression $256w^2 - 676$ can be written as $16w^2 - 26^2$, or $16w16w - 2626$, which factors as $16w - 2616w + 26$.

Choice A is incorrect. This expression is equivalent to $256w^2 - 832w + 676$.

Choice B is incorrect. This expression is equivalent to $64w^2 - 169$.

Choice C is incorrect. This expression is equivalent to $64w^2 - 208w + 169$.

▼ 1.3 — A

Choice A is correct. Since $2$ is a common factor of each of the terms in the given expression, the expression can be rewritten as $2x^2 + 19x + 5$. Therefore, the factors of the given expression are $2$ and $x^2 + 19x + 5$. Of these two factors, only $2$ is listed as a choice.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is a term of the given expression, not a factor of the given expression.

Choice D is incorrect. This is a term of the given expression, not a factor of the given expression.

▼ 1.4 — A

Choice A is correct. The given expression can be rewritten as $8 \cdot 7y \cdot yz \cdot z$, which is equivalent to $56y^2z^2$, or $56y^2z^2$.

Choice B is incorrect. This expression is equivalent to $8yzy7$.

Choice C is incorrect. This expression is equivalent to $8zy7$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.5 — C

Choice C is correct. Applying the distributive property to the given expression yields $d8d^2 - 3 - 68d^2 - 3$. Applying the distributive property once again to this expression yields $d8d^2 + d{-}3 + {-}68d^2 + {-}6{-}3$, or $8d^3 + {-}3d + {-}48d^2 + 18$. This expression can be rewritten as $8d^3 - 48d^2 - 3d + 18$. Thus, $d - 68d^2 - 3$ is equivalent to $8d^3 - 48d^2 - 3d + 18$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.6 — B

Choice B is correct. Each term in the given expression, $100x^2 + 2{,}600$, has a common factor of $100$. Therefore, the expression can be rewritten as $100(x^2) + 100(26)$, or $100(x^2 + 26)$.

Choice A is incorrect. This expression is equivalent to $2{,}600x^2$, not $100x^2 + 2{,}600$.

Choice C is incorrect. This expression is equivalent to $260{,}000x^2$, not $100x^2 + 2{,}600$.

Choice D is incorrect. This expression is equivalent to $260{,}000x^2 + 2{,}600$, not $100x^2 + 2{,}600$.

▼ 1.7 — B

Choice B is correct. The given expression may be rewritten as $x^2 + 8x - 5x - 40$. Since the first two terms of this expression have a common factor of $x$ and the last two terms of this expression have a common factor of $-5$, this expression may be rewritten as $xx + x8 - 5x - 58$, or $xx + 8 - 5x + 8$. Since each term of this expression has a common factor of $x + 8$, it may be rewritten as $x - 5x + 8$.

Alternate approach: An expression of the form $x^2 + bx + c$, where $b$ and $c$ are constants, can be factored if there are two values that add to give $b$ and multiply to give $c$. In the given expression, $b = 3$ and $c = -40$. The values of $-5$ and $8$ add to give $3$ and multiply to give $-40$, so the expression can be factored as $x - 5x + 8$.

Choice A is incorrect. This expression is equivalent to $x^2 + 6x - 40$, not $x^2 + 3x - 40$.

Choice C is incorrect. This expression is equivalent to $x^2 - 3x - 40$, not $x^2 + 3x - 40$.

Choice D is incorrect. This expression is equivalent to $x^2 - 6x - 40$, not $x^2 + 3x - 40$.

▼ 1.8 — B

Choice B is correct. Since $34$ is a common factor of each term in the given expression, the expression can be rewritten as $34x + y$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This expression is equivalent to $34y + 34y$.

Choice D is incorrect. This expression is equivalent to $34x + 34x$.

▼ 1.9 — D

Choice D is correct. Expanding the given expression using the distributive property yields $2a^2(a)+2a^2(3)$. Combining like terms yields $2a^2(a^1)+(2\times3)(a^2)$, or $2a^{2+1}+6a^2$, which is equivalent to $2a^3+6a^2$.

Choices A and B are incorrect and may result from incorrectly combining like terms. Choice C is incorrect and may result from distributing $2a^2$ only to a, and not to 3, in the given expression.

▼ 1.10 — A

Choice A is correct. The expression $4(x^2 + 6)$ can be rewritten as $4x^2 + 4(6)$, which is equivalent to $4x^2 + 24$.

Choice B is incorrect. This expression is equivalent to $4x^2 + \frac{5}{2}$, not $4x^2 + 6$.

Choice C is incorrect. This expression is equivalent to $4x^2 + \frac{3}{2}$, not $4x^2 + 6$.

Choice D is incorrect. This expression is equivalent to $4x^2 - \frac{1}{2}$, not $4x^2 + 6$.

▼ 1.11 — A

Choice A is correct. Since $(x^2-x)$ is a common term in the original expression, like terms can be added: $2(x^2-x)+3(x^2-x)=5(x^2-x)$. Distributing the constant term 5 yields $5x^2-5x$.

Choice B is incorrect and may result from not distributing the negative signs in the expressions within the parentheses. Choice C is incorrect and may result from not distributing the negative signs in the expressions within the parentheses and from incorrectly eliminating the $x^2$-term. Choice D is incorrect and may result from incorrectly eliminating the x-term.

▼ 1.12 — B

Choice B is correct. Since each term in the given expression has a common factor of $x^2$, it can be rewritten as $(13 - 7)x^2$, or $6x^2$. Therefore, the expression $6x^2$ is equivalent to

$13x^2 - 7x^2$.

Alternate approach: Since the two terms of the given expression are both constant multiples of $x^2$, they are like terms and can, therefore, be combined through subtraction. Subtracting like terms in the expression $13x^2 - 7x^2$ yields $6x^2$.

Choice A is incorrect. This expression is equivalent to $(13x)(-7x)$, not $13x^2 - 7x^2$.

Choice C is incorrect. This expression is equivalent to $13x^2 + 7x^2$, not $13x^2 - 7x^2$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.13 — D

Choice D is correct. It's given that $2x^2 + ax$ is equivalent to $x(2x + 7)$ for some constant a. Distributing the x over each term in the parentheses gives $2x^2 + 7x$, which is in the same form as the first given expression, $2x^2 + ax$. The coefficient of the second term in $2x^2 + 7x$ is 7. Therefore, the value of a is 7.

Choice A is incorrect. If the value of a were 2, then $2x^2 + ax$ would be equivalent to $2x^2 + 2x$, which isn't equivalent to $x(2x + 7)$. Choice B is incorrect. If the value of a were 3, then $2x^2 + ax$ would be equivalent to $2x^2 + 3x$, which isn't equivalent to $x(2x + 7)$. Choice C is incorrect. If the value of a were 4, then $2x^2 + ax$ would be equivalent to $2x^2 + 4x$, which isn't equivalent to $x(2x + 7)$.

▼ 1.14 — A

Choice A is correct. The given expression $(2x^2 - 4) - (-3x^2 + 2x - 7)$ can be rewritten as $2x^2 - 4 + 3x^2 - 2x + 7$. Combining like terms yields $5x^2 - 2x + 3$.

Choices B, C, and D are incorrect and may be the result of errors when applying the distributive property.

▼ 1.15 — D

Choice D is correct. In the given expression, the first two terms, $9x^2$ and $7x^2$, are like terms. Combining these like terms yields $9x^2 + 7x^2$, or $16x^2$. It follows that the expression $9x^2 + 7x^2 + 9x$ is equivalent to $16x^2 + 9x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.16 — B

Choice B is correct. Since $2xy$ is a common factor of each term in the given expression, the expression can be rewritten as $2xy8x^2y + 7$.

Choice A is incorrect. This expression is equivalent to $16x^2y^2 + 14xy$.

Choice C is incorrect. This expression is equivalent to $28x^3y^2 + 14xy$.

Choice D is incorrect. This expression is equivalent to $112x^3y^2 + 14xy$.

▼ 1.17 — D

Choice D is correct. The given expression can be rewritten as $9x^3 + 6x^3 + 5x^2 + 5x + 7 - 5$. Combining like terms in this expression yields $15x^3 + 5x^2 + 5x + 2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.18 — D

Choice D is correct. The given expression is equivalent to $9x^2 + 6 - 8x^2$. The terms $9x^2$ and $-8x^2$ are like terms. Combining these like terms yields $x^2$. It follows that the expression $(9x^2 + 6) - (8x^2)$ is equivalent to $x^2 + 6$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is equivalent to $(9x^2 + 6) + (8x^2)$, not the given expression.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.19 — B

Choice B is correct. The given expression can be rewritten as $d^2 + 8 + 3$. Adding 8 and 3 in this expression yields $d^2 + 11$.

Choice A is incorrect. This expression is equivalent to $d^2 + 83$.

Choice C is incorrect. This expression is equivalent to $8 + d^2 - 3$.

Choice D is incorrect. This expression is equivalent to $-8 + d^2 - 3$.

▼ 1.20 — D

Choice D is correct. The expression $2x^3 + 4$ has two terms, $2x^3$ and 4. The greatest common factor of these two terms is 2. Factoring 2 from each of these terms yields $2(x^3) + 2(2)$, or $2(x^3 + 2)$.

Choices A and B are incorrect because 4 is not a factor of the term $2x^3$. Choice C is incorrect and may result from factoring 2 from $2x^3$ but not from 4.

▼ 1.21 — A

Choice A is correct. Since 5 is a factor of both terms, $5x$ and 15, the given expression can be factored and rewritten as $5(x + 3)$.

Choice B is incorrect and may result from subtracting 5 from the constant when factoring 5 from the given expression. Choice C is incorrect and may result from factoring 5 from only the first term, not both terms, of the given expression. Choice D is incorrect and may result from adding 5 to the constant when factoring 5 from the given expression.

▼ 1.22 — A

Choice A is correct. Since each term of the given expression has a factor of $5x$, it can be rewritten as $5xx - 5x10y^2$, or $5xx - 10y^2$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.23 — B

Choice B is correct. Combining like terms inside the parentheses of the given expression, $20w - 4w + 3w$, yields $20w - 7w$. Combining like terms in this resulting expression yields $13w$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.24 — C

Choice C is correct. Since $x$ is a common factor of each term in the given expression, the given expression can be rewritten as $x23x^2 + 2x + 9$.

Choice A is incorrect. This expression is equivalent to $23x^3 + 46x^2 + 207x$.

Choice B is incorrect. This expression is equivalent to $207x^4 + 18x^3 + 9x$.

Choice D is incorrect. This expression is equivalent to $34x^3 + 34x^2 + 34x$.

▼ 1.25 — D

Choice D is correct. Grouping like terms, the given expressions can be rewritten as $r^3 + (5r^2 + r^2) + 8r + (7 + 12)$. This can be rewritten as $r^3 + 6r^2 + 8r + 19$.

Choice A is incorrect and may result from adding the two sets of unlike terms, $r^3$ and $r^2$ as well as $5r^2$ and $8r$, and then adding the respective exponents. Choice B is incorrect and may result from adding the unlike terms $r^3$ and $r^2$ as if they were $r^3$ and $r^3$ and adding the unlike terms $5r^2$ and $8r$ as if they were $5r^2$ and $8r^2$. Choice C is incorrect and may result from errors when combining like terms.

▼ 1.26 — A

Choice A is correct. The expression $19x^2 - 7$ can be rewritten as $19x^2 - 197$, which is equivalent to $19x^2 - 133$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.27 — D

Choice D is correct. The given expression shows addition of two like terms. Therefore, the given expression is equivalent to $50 + 5x^2$, or $55x^2$.

Choice A is incorrect. This expression is equivalent to $505x^2$, not $50 + 5x^2$.

Choice B is incorrect. This expression is equivalent to $\frac{50}{5}x^2$, not $50 + 5x^2$.

Choice C is incorrect. This expression is equivalent to $50 - 5x^2$, not $50 + 5x^2$.

▼ 1.28 — A

Choice A is correct. Since each term of the given expression has a common factor of $w$, it may be rewritten as $w(15w + 8)$. Therefore, the expression $w(15w + 8)$ is equivalent to $15w^2 + 8w$.

Choice B is incorrect. This expression is equivalent to $120w^2 + 8w$, not $15w^2 + 8w$.

Choice C is incorrect. This expression is equivalent to $120w^3 + 15w^2$, not $15w^2 + 8w$.

Choice D is incorrect. This expression is equivalent to $23w^2 + 23w$, not $15w^2 + 8w$.

▼ 1.29 — B

Choice B is correct. The given expression can be rewritten as $11x^3 + -5x^3$. Since the two terms of this expression are both constant multiples of $x^3$, they are like terms and can, therefore, be combined through addition. Adding like terms in the expression $11x^3 + -5x^3$ yields $6x^3$.

Choice A is incorrect. This is equivalent to $11x^3 + 5x^3$, not $11x^3 - 5x^3$.

Choice C is incorrect. This is equivalent to $11x^6 - 5x^6$, not $11x^3 - 5x^3$.

Choice D is incorrect. This is equivalent to $11x^6 + 5x^6$, not $11x^3 - 5x^3$.

▼ 1.30 — C

Choice C is correct. In the given expression, $6x$ and $5x$ are like terms. Combining these like terms yields $11x$. It follows that the expression $6x + 5x + 4y$ is equivalent to $11x + 4y$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.31 — B

Choice B is correct. The expression $16x + 15$ can be rewritten as $16x + 1615$, which is equivalent to $16x + 240$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.32 — B

Choice B is correct. Since $x^3$ is a common factor of each term in the given expression, the expression can be rewritten as $x^3 5x^2 - 6x + 8$.

Choice A is incorrect. This expression is equivalent to $5x^5 - 6x^4$.

Choice C is incorrect. This expression is equivalent to $40x^5 - 48x^4 + 8x^3$.

Choice D is incorrect. This expression is equivalent to $-36x^9 + 48x^8 + 6x^5$.

▼ 1.33 — A

Choice A is correct. Since $x$ is a factor of each term in the given expression, the expression is equivalent to $x9x + x5$, or $x9x + 5$.

Choice B is incorrect. This expression is equivalent to $45x^2 + 5x$, not $9x^2 + 5x$.

Choice C is incorrect. This expression is equivalent to $9x^2 + 45x$, not $9x^2 + 5x$.

Choice D is incorrect. This expression is equivalent to $9x^3 + 5x^2$, not $9x^2 + 5x$.

▼ 1.34 — C

Choice C is correct. Each term in the given expression, $12x + 27$, has a common factor of $3$. Therefore, the expression can be rewritten as $34x + 39$, or $34x + 9$. Thus, the expression $34x + 9$ is equivalent to the given expression.

Choice A is incorrect. This expression is equivalent to $108x + 12$, not $12x + 27$.

Choice B is incorrect. This expression is equivalent to $324x + 27$, not $12x + 27$.

Choice D is incorrect. This expression is equivalent to $27x + 72$, not $12x + 27$.

▼ 1.35 — D

Choice D is correct. For the expression $64t^2s^3 - 56t^3s$, $4$ is a common factor of $64$ and $56$, $t^2$ is a common factor of $t^2$ and $t^3$, and $s$ is a common factor of $s^3$ and $s$. It follows that $4t^2s$ is a common factor of each term in the given expression. Factoring out $4t^2s$ from each term in the expression $64t^2s^3 - 56t^3s$ yields $4t^2s(16s^2 - 14t)$. Therefore, $4t^2s(16s^2 - 14t)$ is equivalent to $64t^2s^3 - 56t^3s$.

Choice A is incorrect. This expression is equivalent to $64ts^3 - 56t^2s^2$, not $64t^2s^3 - 56t^3s$.

Choice B is incorrect. This expression is equivalent to $64t^3s^3 - 56t^2s$, not $64t^2s^3 - 56t^3s$.

Choice C is incorrect. This expression is equivalent to $64t^3s^2 - 56t^2s^2$, not $64t^2s^3 - 56t^3s$.

▼ 1.36 — C

Choice C is correct. Using the distributive property to multiply the terms in the parentheses yields $(2x^3 \cdot x^3) + (2x^3 \cdot -2x) + (3x \cdot x^3) + (3x \cdot -2x)$, which is equivalent to $2x^6 - 4x^4 + 3x^4 - 6x^2$. Combining like terms results in $2x^6 - x^4 - 6x^2$.

Choices A and D are incorrect and may result from conceptual errors when multiplying the terms in the given expression. Choice B is incorrect and may result from adding, instead of multiplying, $(2x^3 + 3x)$ and $(x^3 - 2x)$.

▼ 1.37 — D

Choice D is correct. Expressions in the form $a^2 - b^2$ follow the difference of two squares pattern and can be factored as $(a - b)(a + b)$. In the given expression, $17(x^2 - 100y^2)$, the expression $x^2 - 100y^2$ follows the difference of two squares pattern. It follows that the expression $x^2 - 100y^2$ can be rewritten as $(x - 10y)(x + 10y)$. Therefore, the expression $17(x - 10y)(x + 10y)$ is equivalent to $17(x^2 - 100y^2)$.

Choice A is incorrect. This expression is equivalent to $17(x^2 - 52xy + 100y^2)$, not $17(x^2 - 100y^2)$.

Choice B is incorrect. This expression is equivalent to $17(x^2 + 48xy - 100y^2)$, not $17(x^2 - 100y^2)$.

Choice C is incorrect. This expression is equivalent to $17(x^2 - 20xy + 100y^2)$, not $17(x^2 - 100y^2)$.

▼ 1.38 — B

Choice B is correct. Using the associative and commutative properties of addition, the given expression $(x+5)+(2x-3)$ can be rewritten as $(x+2x)+(5-3)$. Adding these like terms results in $3x+2$.

Choice A is incorrect and may result from adding $(x-5)+(2x+3)$. Choice C is incorrect and may result from adding $(x-5)+(2x-3)$. Choice D is incorrect and may result from adding $(x+5)+(2x+3)$.

▼ 1.39 — D

Choice D is correct. Combining like terms in the given expression yields $9x + 6x + 2y + 3y$, or $15x + 5y$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.40 — B

Choice B is correct. Applying the commutative property of multiplication, the expression $m^4 q^4 z^{-1} mq^5 z^3$ can be rewritten as $m^4 mq^4 q^5 z^{-1} z^3$. For positive values of $x$, $x^a x^b = x^{a+b}$. Therefore, the expression $m^4 mq^4 q^5 z^{-1} z^3$ can be rewritten as $m^{4+1} q^{4+5} z^{-1+3}$, or $m^5 q^9 z^2$.

Choice A is incorrect and may result from multiplying, not adding, the exponents.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.41 — B

Choice B is correct. Applying the distributive property to the given expression yields $2(ab)+2(-3)+2$, or $2ab-6+2$. Adding the like terms $-6$ and 2 results in the expression $2ab-4$.

Choice A is incorrect and may result from multiplying $ab$ by 2 without multiplying $-3$ by 2 when applying the distributive property. Choices C and D are incorrect and may result from computational or conceptual errors.

▼ 1.42 — C

Choice C is correct. The given expression shows subtraction of two like terms. The two terms can be subtracted as follows: $12x^3 - 5x^3 = 12 - 5x^3$, or $7x^3$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the result of adding, not subtracting, the two like terms.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.43 — D

Choice D is correct. The given expression is equivalent to $2x^2 + x + -9 + x^2 + 6x + 1$, which can be rewritten as $2x^2 + x^2 + x + 6x + -9 + 1$. Adding like terms in this expression yields $3x^2 + 7x + -8$, or $3x^2 + 7x - 8$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.44 — D

Choice D is correct. The given expression can also be written as $(x^2+8x+16)^{\frac{1}{4}}$. The trinomial $x^2+8x+16$ can be rewritten in factored form as $(x+4)^2$. Thus, the entire expression can be rewritten as $((x+4)^2)^{\frac{1}{4}}$. Simplifying the exponents yields $(x+4)^{\frac{1}{2}}$.

Choices A, B, and C are incorrect and may result from errors made when simplifying the exponents in the expression $((x+4)^2)^{\frac{1}{4}}$.

▼ 1.45 — A

Choice A is correct. For positive values of $a$, $\frac{a^m}{a^n} = a^{m-n}$, where $m$ and $n$ are integers. Since it's given that $h > 0$ and $q > 0$, this property can be applied to rewrite the given expression as $h^{15-5}q^{7-21}$, which is equivalent to $h^{10}q^{-14}$. For positive values of $a$, $a^{-n} = \frac{1}{a^n}$. This property can be applied to rewrite the expression $h^{10}q^{-14}$ as $h^{10}\frac{1}{q^{14}}$, which is equivalent to $\frac{h^{10}}{q^{14}}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.46 — 9

The correct answer is 9. The given expression can be rewritten as $5x^3 - 3 + -1-4x^3 + 8$. By applying the distributive property, this expression can be rewritten as $5x^3 - 3 + 4x^3 + -8$, which is equivalent to $5x^3 + 4x^3 + -3 + -8$. Adding like terms in this expression yields $9x^3 - 11$. Since it's given that $5x^3 - 3 - -4x^3 + 8$ is equivalent to $bx^3 - 11$, it follows that $9x^3 - 11$ is equivalent to $bx^3 - 11$. Therefore, the coefficients of $x^3$ in these two expressions must be equivalent, and the value of $b$ must be 9.

▼ 1.47 — B

Choice B is correct. Substituting $4k + 5r$ for $a$ and $7k - 12r + 3$ for $b$ in the expression $a - b$ yields $(4k + 5r) - (7k - 12r + 3)$. This expression can be rewritten as $(4k + 5r) + (-1)(7k - 12r + 3)$. By applying the distributive property, this expression can be rewritten as $4k + 5r - 7k + 12r - 3$, which is equivalent to $(4k - 7k) + (5r + 12r) - 3$, or $-3k + 17r - 3$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.48 — B

Choice B is correct. Multiplying $(1 - p)$ by each term of the polynomial within the second pair of parentheses gives $(1 - p)1 = 1 - p$; $(1 - p)p = p - p^2$; $(1 - p)p^2 = p^2 - p^3$; $(1 - p)p^3 = p^3 - p^4$; $(1 - p)p^4 = p^4 - p^5$; $(1 - p)p^5 = p^5 - p^6$; and $(1 - p)p^6 = p^6 - p^7$. Adding these seven expressions together and combining like terms gives $1 + (p - p) + (p^2 - p^2) + (p^3 - p^3) + (p^4 - p^4) + (p^5 - p^5) + (p^6 - p^6) - p^7$, which can be simplified to $1 - p^7$.

Choices A, C, and D are incorrect and may result from incorrectly identifying the highest power of p in the expressions or incorrectly combining like terms.

▼ 1.49 — B

Choice B is correct. For positive values of $a$ and $b$, $a^m b^m = ab^m$, $\sqrt[n]{a} = a^{\frac{1}{n}}$, and $a^{jk} = a^{jk}$. Therefore, the given expression, $\sqrt[7]{x^9 y^9}$, can be rewritten as $\sqrt[7]{xy^9}$. This expression is equivalent to $xy^{9\frac{1}{7}}$, which can be rewritten as $xy^{9 \cdot \frac{1}{7}}$, or $xy^{\frac{9}{7}}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.50 — A

Choice A is correct. Since the given expressions are equivalent and the numerator of the second expression is $\frac{1}{6}$ of the numerator of the first expression, the denominator of the second expression must also be $\frac{1}{6}$ of the denominator of the first expression. By the distributive property, $\frac{1}{6}6x + 42$ is equivalent to $\frac{1}{6}6x + \frac{1}{6}42$, or $x + 7$. Therefore, the value of $b$ is 7.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.51 — ?

The correct answer is 4. It's given that $2x^2+5x-12$ can be rewritten as $(2x-3)(x+k)$; it follows that $(2x-3)(x+k)=2x^2+5x-12$. Expanding the left-hand side of this equation yields $2x^2+2kx-3x-3k=2x^2+5x-12$. Subtracting $2x^2$ from both sides of this equation yields $2kx-3x-3k=5x-12$. Using properties of equality, $2kx-3x=5x$ and $-3k=-12$. Either equation can be solved for k. Dividing both sides of $-3k=-12$ by $-3$ yields $k=4$. The equation $2kx-3x=5x$ can be rewritten as $x(2k-3)=5x$. It follows that $2k-3=5$. Solving this equation for k also yields $k=4$. Therefore, the value of k is 4.

▼ 1.52 — C

Choice C is correct. Multiplying the given functions $f$ and $g$ yields $fx \cdot gx = x^2 + bx9x^2 - 27x$. Applying the distributive property to the right-hand side of this equation yields $fx \cdot gx = x^2 9x^2 - 27x + bx9x^2 - 27x$. Applying the distributive property once again to the right-hand side of this equation yields $fx \cdot gx = x^2 9x^2 - x^2 27x + bx9x^2 - bx27x$, which is equivalent to $fx \cdot gx = 9x^4 - 27x^3 + 9bx^3 - 27bx^2$. Factoring out $x^3$ from the second and third terms yields $fx \cdot gx = 9x^4 + -27 + 9bx^3 - 27bx^2$. Since the left-hand sides of $fx \cdot gx = 9x^4 + -27 + 9bx^3 - 27bx^2$ and $fx \cdot gx = 9x^4 - 26x^3 - 3x^2$ are equal, it follows that $-27 + 9bx^3 = -26x^3$, or $-27 + 9b = -26$, and $-27bx^2 = -3x^2$, or $-27b = -3$. Adding 27 to each side of $-27 + 9b = -26$ yields $9b = 1$. Dividing each side of this equation by 9 yields $b = \frac{1}{9}$. Similarly, dividing each side of $-27b = -3$ by -27 yields $b = \frac{1}{9}$. Therefore, the value of $b$ is $\frac{1}{9}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.53 — C

Choice C is correct. Since each term of the given expression has a common factor of $6x^2y^2$, it may be rewritten as $6x^2y^2(x^6) + 6x^2y^2(2)$, or $6x^2y^2(x^6 + 2)$.

Choice A is incorrect. This expression is equivalent to $12x^8y^2$, not $6x^8y^2 + 12x^2y^2$.

Choice B is incorrect. This expression is equivalent to $6x^6y^2$, not $6x^8y^2 + 12x^2y^2$.

Choice D is incorrect. This expression is equivalent to $6x^6y^2 + 12x^2y^2$, not $6x^8y^2 + 12x^2y^2$.

▼ 1.54 — A

Choice A is correct. The given expression can be rewritten as $(2x+5)^2+(-1)(x-2)+2(x+3)$. Applying the distributive property, the expression $(-1)(x-2)+2(x+3)$can be rewritten as $-1(x)+(-1)(-2)+2(x)+2(3)$, or $-x+2+2x+6$. Adding like terms yields $x+8$. Substituting $x+8$ for $(-1)(x-2)+2(x+3)$ in the given expression yields $(2x+5)^2+x+8$. By the rules of exponents, the expression $(2x+5)^2$ is equivalent to $(2x+5)(2x+5)$. Applying the distributive property, this expression can be rewritten as $2x(2x)+2x(5)+5(2x)+5(5)$, or $4x^2+10x+10x+25$. Adding like terms gives $4x^2+20x+25$. Substituting $4x^2+20x+25$ for $(2x+5)^2$ in the rewritten expression yields $4x^2+20x+25+x+8$, and adding like terms yields $4x^2+21x+33$.

Choices B, C, and D are incorrect. Choices C and D may result from rewriting the expression $(2x+5)^2$ as $4x^2+25$, instead of as $4x^2+20x+25$. Choices B and C may result from rewriting the expression $-(x-2)$ as $-x-2$, instead of $-x+2$.

▼ 1.55 — B

Choice B is correct. Since $\frac{12}{12} = 1$, multiplying the exponent of the given expression by $\frac{12}{12}$ yields an equivalent expression: $a^{\frac{11}{12}\frac{12}{12}} = a^{\frac{132}{144}}$. Since $\frac{132}{144} = 132\frac{1}{144}$, the expression $a^{\frac{132}{144}}$ can be rewritten as $a^{132\frac{1}{144}}$. Applying properties of exponents, this expression can be rewritten as $a^{132\,\frac{1}{144}}$. An expression of the form $m^{\frac{1}{k}}$, where $m > 0$ and $k > 0$, is equivalent to $\sqrt[k]{m}$. Therefore, $a^{132\,\frac{1}{144}}$ is equivalent to $\sqrt[144]{a^{132}}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.56 — B

Choice B is correct. An expression in the form $x^{\frac{1}{k}}$, where $x > 0$ and $k > 0$, is equivalent to $\sqrt[k]{x}$. It follows that $x^{\frac{1}{14}}$, where $x > 0$, is equivalent to $\sqrt[14]{x}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.57 — D

Choice D is correct. Taking the square root of an exponential expression halves the exponent, so $\frac{\sqrt{16x^4y^8}}{x^3} = \frac{4x^2y^4}{x^3}$, which further reduces to $\frac{4y^4}{x}$. This can be rewritten as $4x^{-1}y^4$.

Choice A is incorrect and may result from neglecting the denominator of the given expression and from incorrectly calculating the square root of 16. Choice B is incorrect and may result from rewriting $\frac{1}{x}$ as $x^1$ rather than $x^{-1}$. Choice C is incorrect and may result from taking the square root of the variables in the numerator twice instead of once.

▼ 1.58 — B

Choice B is correct. The expression $x^2 + 11^2$ can be written as $x^2 + 11x^2 + 11$, which is equivalent to $x^2 x^2 + 11 + 11x^2 + 11$. Distributing $x^2$ and 11 to $x^2 + 11$ yields $x^4 + 11x^2 + 11x^2 + 121$, or $x^4 + 22x^2 + 121$. The expression $x - 5x + 5$ is equivalent to $x - 5x + x - 55$. Distributing $x$ and 5 to $x - 5$ yields $x^2 - 5x + 5x - 25$, or $x^2 - 25$. Therefore, the expression $x^2 + 11^2 + x - 5x + 5$ is equivalent to $x^4 + 22x^2 + 121 + x^2 - 25$, or $x^4 + 22x^2 + 121 + x^2 - 25$. Combining like terms in this expression yields $x^4 + 23x^2 + 96$.

Choice A is incorrect. Equivalent expressions must be equivalent for any value of $x$. Substituting $0$ for $x$ in this expression yields $-14$, whereas substituting $0$ for $x$ in the given expression yields $96$.

Choice C is incorrect. Equivalent expressions must be equivalent for any value of $x$. Substituting $0$ for $x$ in this expression yields $121$, whereas substituting $0$ for $x$ in the given expression yields $96$.

Choice D is incorrect. Equivalent expressions must be equivalent for any value of $x$. Substituting $0$ for $x$ in this expression yields $146$, whereas substituting $0$ for $x$ in the given expression yields $96$.